

- 1(a) Zac is planning to write a report on the music preferences of the students at his college. There is a large number of students at the college.

State one reason why Zac might wish to obtain information from a sample of students, rather than from all the students. [1]

- (b) Amaya suggests that Zac should use a sample that is stratified by school year.

Give one advantage of this method as compared with random sampling, in this context. [1]

- (c) Zac decides to take a random sample of 60 students from his college. He asks each student how many hours per week, on average, they spend listening to music during term. From his results he calculates the following statistics.

Mean	Standard deviation	Median	Lower quartile	Upper quartile
21.0	4.20	20.5	18.0	22.9

Sundip tells Zac that, during term, she spends on average 30 hours per week listening to music.

Discuss briefly whether this value should be considered an outlier. [3]

- (d) Layla claims that, during term, each student spends on average 20 hours per week listening to music. Zac believes that the true figure is higher than 20 hours. He uses his results to carry out a hypothesis test at the 5% significance level.

Assume that the time spent listening to music is normally distributed with standard deviation 4.20 hours.

Carry out the test.

[7]

2(a) Jane conducted a survey. She chose a sample of people from three towns, A, B and C. She noted the following information.

400 people were chosen.

230 people were adults.

55 adults were from town A.

65 children were from town A.

35 children were from town B.

150 people were from town B.

Complete the two-way frequency table. [2]

		Town			Total
		A	B	C	
adult					
child					
Total					

(b) One of the people is chosen at random.

(i) Find the probability that this person is an adult from town A.

[1]

(ii) Given that the person is from town A, find the probability that the person is an adult.

[1]

- (c) For another survey, Jane wanted to choose a random sample from the 820 students living in a particular hostel. She numbered the students from 1 to 820 and then generated some random numbers on her calculator.

The random numbers were 0.114 287 562 and 0.081 859 817.

Jane's friend Kareem used these figures to write down the following sample of five student numbers.

114, 142, 428, 287 and 756

Jane used the same figures to write down the following sample of five student numbers.

114, 287, 562, 81 and 817

- (i) State, with a reason, which one of these samples is not random.

[1]

- (ii) Explain why Jane omitted the number 859 from her sample.

[1]

- 3 A certain forest contains only trees of a particular species. Dipak wished to take a random sample of 5 trees from the forest. He numbered the trees from 1 to 784. Then, using his calculator, he generated the random digits 14 781 049. Using these digits, Dipak formed 5 three-digit numbers. He took the first, second and third digits, followed by the second, third and fourth digits and so on. In this way he obtained the following list of numbers for his sample.

147 478 781 104 49

- (a) Explain why Dipak omitted the number 810 from his list. [1]

- (b) Explain why Dipak's sample is not random. [1]

The mean height of all trees of this species is known to be 4.2 m. Dipak wishes to test whether the mean height of trees in the forest is less than 4.2 m. He now uses a correct method to choose a random sample of 50 trees and finds that their mean height is 4.0 m. It is given that the standard deviation of trees in the forest is 0.8 m.

- (c) Carry out the test at the 2% significance level. [7]

4 Jo is investigating the popularity of a certain band amongst students at her school. She decides to survey a sample of 100 students.

(a) State an advantage of using a stratified sample rather than a simple random sample.

[1]

(b) Explain whether it would be reasonable for Jo to use her results to draw conclusions about all students in the UK.

[1]

- 5 A researcher is investigating the actual lengths of time that patients spend at their appointments with the doctors at a certain clinic. There are 12 doctors at the clinic, and each doctor has 24 appointments per day. The researcher plans to choose a sample of 24 appointments on a particular day.

(a) The researcher considers the following two methods for choosing the sample.

Method A: Choose a random sample of 24 appointments from the 288 on that day.

Method B: Choose one doctor's 1st and 2nd appointments. Choose another doctor's 3rd and 4th appointments and so on until the last doctor's 23rd and 24th appointments.

For each of A and B state a disadvantage of using this method.

[2]

Appointments are scheduled to last 10 minutes. The researcher suspects that the actual times that patients spend are more than 10 minutes on average. To test this suspicion, he uses method A, and takes a random sample of 24 appointments. He notes the actual time spent for each appointment and carries out a hypothesis test at the 1% significance level.

(b) Explain why a 1-tail test is appropriate.

[1]

The population mean of the actual times that patients spend at their appointments is denoted by μ minutes.

(c) Assuming that $\mu = 10$, state the probability that the conclusion of the test will be that μ is not greater than 10.

[1]

The actual lengths of time, in minutes, that patients spend for their appointments may be assumed to have a normal distribution with standard deviation 3.4.

(d) Given that the total length of time spent for the 24 appointments is 285 minutes, carry out the test. [7]

(e) In part (d) it was necessary to use the fact that the sample mean is normally distributed. Give a reason why you know that this is true in this case. [1]

6 The governors of a school wish to investigate the opinions of the parents and guardians of the pupils. The secretary of the governors distributes a questionnaire to all parents and guardians who are present at a particular Parents' Evening.

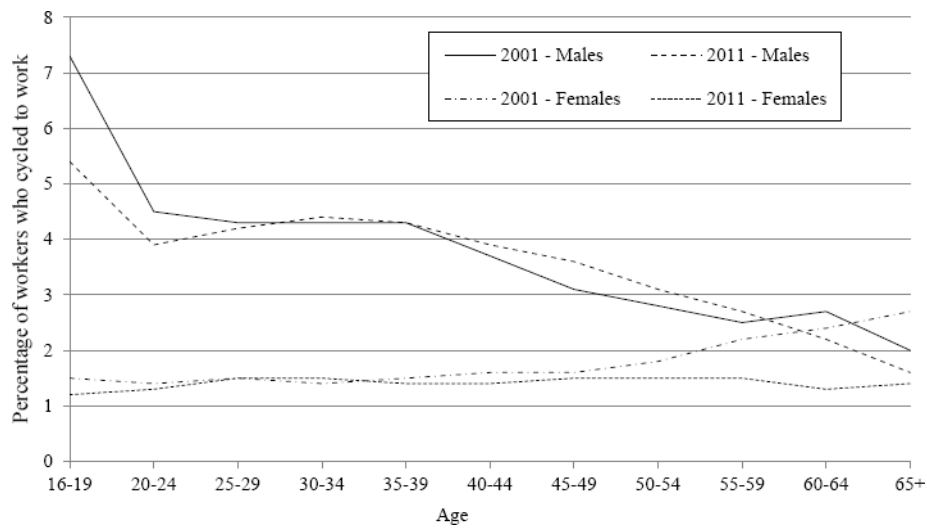
(i) Explain why this method of sampling may not give reliable results.

[1]

(ii) Suggest a better method of sampling, using random numbers.

[2]

7 The diagram below shows some "Cycle to work" data taken from the 2001 and 2011 UK censuses. The diagram shows the percentages, by age group, of male and female workers in England and Wales, excluding London, who cycled to work in 2001 and 2011.



The following questions refer to the workers represented by the graphs in the diagram.

- (a) A researcher is going to take a sample of men and a sample of women and ask them whether or not they cycle to work. Why would it be more important to stratify the sample of men? [1]
- (b) A research project followed a randomly chosen large sample of the group of male workers who were aged 30-34 in 2001. Does the diagram suggest that the proportion of this group who cycled to work has increased or decreased from 2001 to 2011? Justify your answer. [2]
- (c) Write down one assumption that you have to make about these workers in order to draw this conclusion. [1]

- 8 A club secretary wishes to survey a sample of members of his club. He uses all members present at a particular meeting as his sample.

(a) Explain why this sample is likely to be biased.

[1]

Later the secretary decides to choose a random sample of members. The club has 253 members and the secretary numbers the members from 1 to 253. He then generates random 3-digit numbers on his calculator. The first six random numbers generated are 156, 965, 248, 156, 073 and 181. The secretary uses each number, where possible, as the number of a member in the sample.

(b) Find possible numbers for the first four members in the sample.

[2]

- 9 It is required to select a random sample of 30 pupils from a school with 853 pupils. A student suggests the following method.

“Give each pupil sequentially a three-digit number from 001 to 853. Use a calculator to generate random three-digit numbers from 0.000 to 0.999 inclusive, multiply the answer by 853, add 1 and round off to the nearest whole number. Select the corresponding pupil, and repeat as necessary.”

(i) Determine which pupil would be picked for each of the following calculator outputs:

0.103, 0.104, 0.105, 0.106, 0.107.

[2]

(ii) Use your answers to part (i) to show that this method is biased, and suggest an improvement.

[2]

END OF QUESTION PAPER

Mark Scheme

Question			Answer/Indicative content	Marks	Part marks and guidance	
1	a		Population large oe	B1 [1]	or, eg Would take too long to contact all students. NOT "Easier"	
	b		eg: Includes students from all years (or ages) Numbers in years in correct proportions Different years might like different music	B1 [1]	or: Different years may have different numbers of students NOT "It's more representative" or "Takes all students into account" "You get a range of people" "It avoids bias"	
	c		$21 + 2 \times 4.2 = 29.4$ $22.9 + 1.5(22.9 - 18.0) = 30.25$ Unclear whether 30 is an outlier	B1 B1 B1 [3]	Allow "30 is more than 2 sds away from the mean" Allow "30 is less than $1.5 \times \text{IQR}$ from UQ" or eg "It depends which definition you use." Any comment implying uncertainty Ignore comments about $\text{mean} \pm 3 \text{ sds}$ or $\text{mean} \pm 1 \text{ sd}$ <u>Examiner's Comments</u> The specification (paragraph 2.02j) gives two definitions for an outlier. Most candidates used only one of these definitions, and so did not discover that they give different results. What was required for full marks was a recognition of the uncertainty about whether or not 30 should be considered an outlier. Candidates who used other definitions (such as $x > \mu + 3s$ or $x > \mu + s$) gained no credit. Some candidates did a calculation that mixed the median with the standard deviation or the mean with the IQR.	

Mark Scheme

Question		Answer/Indicative content	Marks	Part marks and guidance	
	d	$H_0: \mu = 20$ $H_1: \mu > 20$ where $\mu = \text{pop}$ $\bar{X} \sim N(20, \frac{4.2^2}{60})$ and $\bar{X} = 21$ $P(\bar{X} > 21) = 0.0326$ Compare 0.05 Alternative methods for M1A1A1 or $\frac{a-20}{4.2/\sqrt{60}} = 1.645$ ($a = 20.9$) CV = 20.9 $21 > 20.9$ or 21 not in acceptance region or $\frac{21-20}{4.2/\sqrt{60}} (= 1.84)$ $z_{\text{calc}} = 1.84$ Compare 1.645 $\bar{X} \sim N(20, \frac{4.2^2}{60})$ and $\bar{X} = \frac{1260}{60}$ or 21 $P(\bar{X} < 21) \leq 0.9674$ Compare 0.95 Reject H_0 Condon Accept H_1 There is evidence that (mean) time spent is > 20 hours or eg there is evidence to support Zac's belief NB Use of $4.2^2/60$ as sd gives $p = 0.000335$	B1 B1 M1 A1 A1 Alternative methods for M1A1A1 M1 A1 A1 M1 A1 A1 M1 A1 A1 M1 A1f [7]	NB Allow 2 sf throughout Allow other letters, not X unless defined. Not $\bar{X}$ B1B0 for 1 error eg 2-tail or: 2-tail B1B0 $\mu = \text{sa}$ B1B0 mple mean i mplied undefi B1B0 Not incB1B0 ned μ lude value 20 not in B1B0 eg H_0 B1B0 terms = 20 of para etc: meter Correct distribution and value of $\bar{X}$ stated or implied eg by 0.0326 or 0.967 or 20.9 or 1.84 or 0.000335 even if within incorrect statement eg $P(X = 21) =$ 0.0326 Condone $\frac{4.2^2}{\sqrt{60}}$ or $\frac{4.2^2}{60^2}$ or $\frac{4.2}{60}$ BC Allow 2 sf, ie 0.033 Dep 0.0326 or 1.84 or 0.9674 or $P(X > 21)$ or $P(\geq 21)$ soi Must compare like with like, eg NOT prob cf z-value or large prob cf small prob or CV cf wrong end of acceptance region Alternative methods for M1A1A1 Condone $\frac{4.2^2}{\sqrt{60}}$ or $\frac{4.2^2}{60^2}$ or	

Mark Scheme

Question			Answer/Indicative content	Marks	Part marks and guidance	
					$\frac{4.2}{60}$ Condone $\frac{4.2^2}{\sqrt{60}}$ or $\frac{4.2^2}{60^2}$ or $\frac{4.2}{60}$ Condone $\frac{4.2^2}{\sqrt{60}}$ or $\frac{4.2^2}{\sqrt{60}}$ or $\frac{4.2}{60}$ BC Dependent on clearly valid comparison of like with like. Dep 0.0485 or 0.951 or 20.9 or 1.84 or $P(X > 21)$ or $P(X \geq 21)$ soi May be implied by conclusion, eg "There is evidence that mean time is > 20 hours" M1A1 In context, not definite; eg "Mean time is > 20 hours": A0 But "There is evidence to reject H0 and that mean time is > 20 h" Allow opposite conclusion, ft their values, if above conditions met and loses 2nd A1. But potentially can score all the other 6 marks <u>Examiner's Comments</u>  Misconception One difficult feature of hypothesis testing is that in some cases the calculated value of the statistic is significant when it is greater than the critical value, whereas in other cases the calculated value is	

Mark Scheme

Question			Answer/Indicative content	Marks	Part marks and guidance	
					significant when it is less than the critical value. In both these questions there was confusion about this. Many candidates used the sample mean, 21, as the mean of the distribution, rather than the suggested population mean of 20.	
			Total	12		
2	a		$\begin{array}{ccc ccc} (55) & 115 & (60) & & (230) & \\ (65) & (35) & 70 & & 170 & \\ \hline 120 & (150) & 130 & & (400) & \end{array}$	B1 B1 [1]	B1 for 10 correct numbers	
	b	i	$\frac{55}{400}$ ISW or $\frac{11}{80}$ or 0.1375 or 0.138 (3 sf)	B1 [1]	NB If figures for A in the table are correct, but those for B and C are wrong, correct answers to parts (b)(i), (ii) can be obtained correctly.	
		ii	$\frac{55}{120}$ ISW or $\frac{11}{24}$ or 0.458 (3 sf)	B1 [1]		
	c	i	Kareem because, eg the 2nd digit of each no. is the same as the 1st digit of the next	B1 [1]	oe. Allow Kareem because the numbers are not independent.	
		ii	Only 820 students or $850 > 820$	B1 [1]	oe	
			Total	6		

Mark Scheme

Question			Answer/Indicative content	Marks	Part marks and guidance		
3		a	Only 784 trees and 810 > 784	E1 (AO 2.4) [1]	or other similar		
		b	eg Each no. not independent of previous no. Each no. is related to the next	E1 (AO 2.3) [1]	Allow 2nd digit of each no. is 1st of next Consecutive nos share two digits Ignore all else	or similar correct Digits are re-used	

Examiner's Comments

This question was very well answered.

Examiner's Comments

Only a minority of candidates saw the point. Some suggested reasons such as "Only 3-digit numbers can be selected", "1- or 2-digit numbers are less likely than 3-digit", "It's systematic, not random", "He picked the numbers using a pattern", "He only used these digits so, for example, tree number 324 could not be chosen", "The calculator does not give random numbers", "He chose which numbers to use", Many gave that old favourite fall-back answer (not appropriate here): "Every tree does not have an equal chance of being selected".

Mark Scheme

Question			Answer/Indicative content	Marks	Part marks and guidance		
		c	$H_0: \mu = 4.2$ $H_1: \mu < 4.2$ where μ is mean height of trees (in the wood) $\bar{X} \sim N(4.2, \frac{0.8^2}{50})$ an $\bar{X} < 4.0$ or $\bar{X} \leq 4.0$ d $P(\bar{X} < 4.0) = 0.038549...$ or 0.039 Compare 0.02 Do not reject H_0 There is insufficient evidence that mean height of these trees in the wood is less than 4.2m.	B1 (AO 1.1) B1 (AO 2.5) M1 (AO 3.3) A1 (AO 3.4) A1 (AO 1.1) M1 (AO 2.2b) A1f (AO 3.5a) [7]	Allow other letters except X or $\bar{X}$ One error, eg undefined μ or 2-tail: B0B1 Stated or implied Allo $\bar{X} > 4.0$ or w $\bar{X} = 4.0$ BC Allow 0.038 NB 0.038... implies M1A1 dep $P(\bar{X} < 4.0)$ attempted Allow Accept H_0 dep $P(\bar{X} < 4.0)$ attempted In context, not definite; eg "Mean height not less than 4.2m": A0 Examiner's Comments This question was well answered by many candidates. Perhaps the most common error was the omission of a definition of μ in the hypotheses. Some candidates made errors in	$\Phi^{-1}(0.98)$ (= 2.054) $4.2 - 2.054 \times \frac{0.8}{\sqrt{50}}$ (= 3.968) comp their 3.968 with 4.0 Can be implied by conclusion	

Mark Scheme

Question			Answer/Indicative content	Marks	Part marks and guidance	
					the standard deviation of $\bar{X}$, such as $\frac{0.8}{50}$. A few candidates gave a "definite" conclusion, such as "The mean height is not less than 4.2". These candidates lost the final mark. A few others omitted to give a conclusion in context, just stating, "There is insufficient evidence to reject H_0". Some candidates used the "critical value" method, comparing the c.v. with the relevant value of z. This method is, of course, acceptable, but with the advent of more powerful calculators, the method which finds the relevant probability (here $P(\bar{X} < 4.0)$) and compares this with the significance level is quicker and simpler.	
			Total	9		

Mark Scheme

Question			Answer/Indicative content	Marks	Part marks and guidance		
4		a	Any mention of diff categories or types, eg Includes students in all years (or classes) More representative of diff ages Prevents obtaining too many in one year gp	E1 (AO2.4) [1]	or any mention of proportions , eg Uses the right proportions of students Ignore all else	NOT eg: Wider variety results More representative of pop	
Examiner's Comments Many incorrect answers were seen, such as "It is more accurate" or "Not biased" or "It's easier" or "With stratified, she could ask people who were interested". Many inadequate answers were also seen, such as "It is more representative" or "More reliable". It is important to refer to the context and not simply provide a generic statement.							

Mark Scheme

Question			Answer/Indicative content	Marks	Part marks and guidance		
		b	Must include <u>reason</u> why not rep've, eg Her school may be biased Students' friends may be in local band Opinions at one school not indep of one another Ignore all else	E1 (AO2.3) [1]	except allow Small sample or Only 100 students in sample Student opinion elsewhere may differ Diff conditions, or diff types of school, in diff areas, Diff bands liked in diff areas Examiner's Comments Here many inadequate answers were seen, such as "Her school might not be representative of schools in the UK". Some hint of a reason why the school might not be representative was required for the mark.	NOT: Not representative of UK "No" may be implied	
			Total	2			

Mark Scheme

Question			Answer/Indicative content	Marks	Part marks and guidance		
5		a	Method A: eg May not be representative, e.g. may have many (or few) appointments from one Dr Or may have many a.m. appts & few p.m. Method B: eg Time of 1st Dr's 1st & 2nd appointments may not be typical of his later ones. Similar for other Drs	B1(AO 3.5b) B1(AO 3.5b) [2]	Or may not have a good spread of appts Or other sensible Or other sensible		
		b	Researcher suspects "more than" 10 mins	B1(AO 2.4) [1]	oe		
		c	0.99	B1(AO 1.2) [1]			

Mark Scheme

Question			Answer/Indicative content	Marks	Part marks and guidance		
		d	$H_0: \mu = 10$ where μ is pop mean appointment time $H_1: \mu > 10$ $\bar{X} \sim N(10, \frac{3.4}{\sqrt{24}})$ and $X = \frac{285}{24}$ (= 11.875) $P(\bar{X} > 11.875) = 0.00345$ or better Compare with 0.01 Reject H_0 Sufficient evidence that mean time is > 10 mins	B1(AO 1.1) B1(AO 2.5) M1(AO 3.3) A1(AO 3.4) M1(AO 1.1) M1(AO 2.2b) A1(AO 3.5a) [7]	One error, eg undefined μ or 2-tail: B0B1 May be implied or 0.003 BC In context. Not definite, eg "Mean time is > 10 mins": A0		
		e	Times spent assumed normally distributed, hence sample mean also normally distributed	B1(AO 1.2) [1]	oe		
			Total	12			

Mark Scheme

Question			Answer/Indicative content	Marks	Part marks and guidance	
6		i	Biased against those not at the parents' evening	B1 1	Reason for being biased or unrepresentative, needs more than "not all will be at the meeting", e.g. "not all will return the questionnaire" or "those at the meeting may have different opinions" "Biased" can be implied by the reason. Not <i>just</i> "not random" or "not representative", but allow "self-selecting". Ignore irrelevances (e.g. "small sample", but withhold if definitely wrong comment seen). Examiner's Comments The question asked for an explanation of why a method was unsuitable. The following common answers are inadequate for the reasons stated: • "The method is not random": some non-random methods (for instance, a systematic sample with non-random choice of the starting point) can be perfectly adequate. • "The sample is not representative": simple random samples selected without bias may not be representative purely by chance. • "Not everyone would be	

Mark Scheme

Question			Answer/Indicative content	Marks	Part marks and guidance	
					able to reply": <i>any</i> sampling method involves not getting replies from all the population. Some sort of reason for bias was required, such as "those attending the meeting may have stronger opinions". In fact the main problem with any method that asks respondents to return a questionnaire is always that of the "self-selecting sample" (those with stronger opinions are more likely to reply), but few candidates focussed on this problem.	

Mark Scheme

Question			Answer/Indicative content	Marks	Part marks and guidance		
		ii	Obtain list of parents / pupils & number it 1 to n Select using random numbers, ignoring repeats / numbers outside range	B1 B1 2	Number (a list of) parents (sequentially) (statements in brackets can be implied) Mention use of RNs, as <i>only</i> method, <i>and</i> either “ignore repeats” or “ignore outside range” (allow “use RNs in range”)	SC: Allocate <i>random</i> numbers: max B1 unless <i>sorted</i> <i>Not</i> “select numbers randomly” <i>Not</i> hat / lottery machine [RNs required by question] Allow systematic provided random start	
					Examiner's Comments Candidates were required to explain a method involving random numbers, so those who suggested putting names into a hat did not gain full marks. Some candidates said “number the parents randomly”, which is not an appropriate method unless they are then <i>sorted</i> by those random numbers. Some demonstrated a lack of understanding of random numbers by saying “put the numbers into a random number generator”. As in previous years, candidates had to refer to “ignoring repeats” or “ignoring numbers outside the range” in order to score full marks.		

Mark Scheme

Question			Answer/Indicative content	Marks	Part marks and guidance	
			Total	3		
7		a	e.g. From the data given, the proportions of men who cycle to work show much more variability than women, with greater proportions of younger men cycling than older men.	E1(AO2.4) [1]		
		b	The proportion decreased e.g. These workers were in the 40-44 group in 2011, which is a smaller proportion of the population than the 30-34 group in 2001.	B1(AO2.2a) B1(AO2.2b) [2]		
		c	e.g. The age group is still approximately the same size in 2011 Very few (or no) males in this age group join the workforce between 2001 and 2011 Very few (or no) males in this age group leave the workforce between 2001 and 2011 The overall size of the workforce in this age group has not changed much The sample is representative of the whole population	B1(AO2.2b) [1]	For any relevant assumption	
			Total	4		

Mark Scheme

Question			Answer/Indicative content	Marks	Part marks and guidance		
8		a	E.g. Members who attend may be of a particular type E.g. Absent members cannot be included	B1 (AO2.5) [1]	Any correct explanation Sample is not random B0		
		b	156, 248 73, 181	B1 (AO1.1) B1 (AO1.1) [2]	Allow 073	965 must be discarded In <i>this</i> context do not accept a repeat of 156	
			Total	3			

Mark Scheme

Question			Answer/Indicative content	Marks	Part marks and guidance	
9		i	89, 90, 91, 91, 92	B2	All correct; B2; one error (e.g. all – 1), B1	Allow 088, etc Examiner's Comments Most got this right, though some failed to add 1 to the numbers (answering “88, 89, 90, 90, 91”) and some rushed to the conclusion that the numbers always went up by 1.
		ii	Not all equally likely (91 more than 90 etc)	B1	Imply different likelihood / probability	<i>Not “same pupil is selected twice”</i>
		ii	Multiply by 1000 and ignore if > 853	B1	Or equivalent method. Not “ignore repeats”. Ignore extras.	Examiner's Comments Quite a few candidates showed that they had misunderstood the words “which candidates <i>would be picked</i> ”, by answering that the same pupils <i>had been picked</i> twice and that repeats should be ignored. All that was needed was the comment that not all pupils were equally likely to be selected, and that the random numbers should be multiplied by 1000, rejecting numbers greater than 853.
			Total	4		